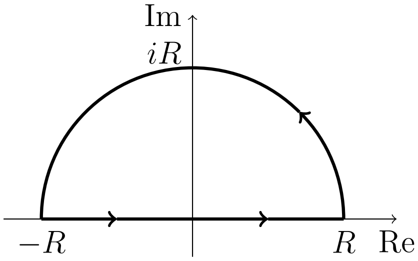
3 Complex Analysis 2024
Tutorial 7

A/Prof Elena Berdysheva
Mita Ramabulana

Cauchy's integral formula

Problem 28 Consider the path γ indicated on the figure below, with $R > 1$:



We denote by γ_1 the segment $[-R, R]$, and by γ_2 the arc.

(a) Prove that $\displaystyle\int_\gamma \frac{1}{1+z^2}\,dz = \pi$ and that $\displaystyle\lim_{R\to\infty}\left|\int_{\gamma_2} \frac{1}{1+z^2}\,dz\right| = 0$.

(b) Conclude that $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \lim_{R\to\infty}\int_{-R}^{R} \frac{1}{1+x^2}\,dx = \pi$.

Problem 29 Evaluate the following integrals.

(a) $\displaystyle\int_\gamma \frac{z^7+1}{z^2(z^4+1)}\,dz$, where $\gamma(t) = 2 + e^{it}$, $t \in [0, 2\pi]$.

(b) $\displaystyle\int_\gamma \frac{e^{-z}}{(z+2)^3}\,dz$, where $\gamma(t) = 3e^{it}$, $t \in [0, 2\pi]$.

(c) $\displaystyle\int_\gamma \frac{1}{(z-1)(z-2)^2}\,dz$, where $\gamma(t) = 2 + \frac{1}{2}e^{it}$, $t \in [0, 2\pi]$.

(d) $\displaystyle\int_\gamma \frac{z^4 e^z}{(z-1)^4}\,dz$, where $\gamma(t) = 2e^{it}$, $t \in [0, 2\pi]$.

Problem 30 Let G be a region, $z_0 \in G$, $r > 0$ such that the closed disc $\overline{D}(z_0; r) \subseteq G$. Denote by γ the circle with center z_0, radius r and positive orientation. Let f be a holomorphic function in G. Calculate

$$\frac{1}{2\pi i}\int_\gamma \frac{f(w)-f(z)}{(w-z)^2}\,dw$$

for $z \in D(z_0; r)$.

Problem 31 Let f be a holomorphic function in the disc $D(0;1) = \{z \in \mathbb{C} : |z| < 1\}$ such that $|f(z)| \leq 1$ for all $z \in D(0;1)$. Show that

$$|f'(z)| \leq 4 \quad \text{for } |z| \leq \frac{1}{2}.$$